Let 3 : Root test, alternating series, Leibnitz test.

Cauchy's n-th root test : If $\sum a_n$ is a +ve term series ~~then~~ &

$\lim_{n\to\infty} (a_n)^{1/n} = l$ then $l<1 \to$ conv, $l>$ div, $l=1$ test fails.

Eg. 1) $\sum \left(1+\frac{1}{n}\right)^{-n^2}$ $\lim_{n\to\infty} (a_n)^{1/n} = \lim_{n\to\infty} \left(1+\frac{1}{n}\right)^{-n} = \frac{1}{e} < 1$ cgt.

2) ~~$\sum x$~~ For what values of $x \geq 0$, $\sum \frac{x^n}{n}$ converges or diverges?

$\lim_{n\to\infty} (a_n)^{1/n} = \lim_{n\to\infty} \frac{x}{n^{1/n}} = x$ $\because \lim_{n\to\infty} n^{1/n} = 1$

$\therefore$ converges when $0<x<1$ & at $x=0$, the sum is 0 $\therefore$ for $0 \leq x < 1$

div when $x \geq 1$ $\because$ at $x=1$ we get $\sum \frac{1}{n}$ which is div.

Oscillating finitely & infinitely series:

old friend: $\sum_{n=1}^{\infty} r^n$ $\therefore$ at $r=-1$ $S_n = \begin{cases} 0 & n \text{ is even} \\ -1 & n \text{ is odd} \end{cases}$ $\lim S_n = \{0,-1\}$ oscill. finitely

Other way: bdd + not conv $\to$ osci. finitely (previous lectures)

Why not conv ?? bcz not monotonic (but bdd)

Again for $r=-2$ $\sum r^n = \sum (-2)^n = \sum (-1)^n 2^n$

$S_n = \frac{r(1-r^n)}{1-r}$, at $r=-2$, $S_n = \frac{-2[1-(-2)^n]}{3}$ Look at $1-(-2)^n$

It is ~~as~~ unbdd + not div $\longrightarrow$ oscillating infinitely (previous lecture)

& Hence, the series $\sum r^n$ at $r=-2$ osci. infinitely.

Observation: ~~Not of +ve terms~~ They are infinite series but NOT of +ve terms. & that brings us to the notion of $\to$

Alternating series: If $a_n > 0 \forall n$, then $\sum (-1)^n a_n$ is alternating series.

Do all alternating series either oscillate finitely or infinitely?

False Eg. $\sum_{n=1}^{\infty} \left(-\frac{1}{4}\right)^n = \frac{-1/4}{1-(-\frac{1}{4})} = \frac{1}{5}$ (conv.)

Leibnitz's test: If a_n is mono. ↓ing & $\lim a_n = 0$ then $\sum (-1)^n a_n$ converges.

Eg. 1) $\sum \frac{(-1)^n}{n}$ Here $a_n = \frac{1}{n}$ mono ↓ing & $\lim \frac{1}{n} = 0$

Hence, by Leibnitz test, $\sum \frac{(-1)^n}{n}$ converges.

2) $\sum \frac{(-1)^n}{\log n}$ Here $a_n = \frac{1}{\log n}$ Now $\because n+1 > n$

$\log(n+1) > \log n$

& $\frac{1}{\log n} > \frac{1}{\log(n+1)}$

$a_n > a_{n+1}$ mono ↓ing

& $\lim \frac{1}{\log n} = 0$

by Leibnitz test, $\sum \frac{(-1)^n}{\log n}$ conv.

3) $\sum (-1)^n \left(1+\frac{1}{n}\right)$ Here $a_n = 1+\frac{1}{n}$ mono ↓ing but $\lim a_n \neq 0$

Leibnitz test can be applied.

(4) $\sum \frac{(-1)^n}{n^2}$ a_n mono ↓ing & $\lim a_n = 0$ Hence conv.

Now, look at Eg (1) & (4). in (1) the series $\sum a_n = \sum \frac{1}{n}$ diverges but its alternating companion converges, while in (4) $\sum a_n = \sum \frac{1}{n^2}$ converges & its alternating companion also converges. Let us assign some nomenclature to the series exhibiting such phenomenon, namely the conditionally con & abs. conv. series.

(1) If $\sum a_n$ is a series of real no. & if $\sum |a_n|$ is cgt, we say $\sum a_n$ is abs. cgt.

(2) If $\sum |a_n|$ is div. but $\sum a_n$ is cgt, we say $\sum a_n$ is cond. cgt.

Eg $\sum_{n=1}^{\infty} (-1)^n \frac{(\log n)^a}{n}$. for what values of 'a', this series is abs & cond. cgt.

For $a = -2$, we get

Eg. 1) $\sum (-1)^n \sin \frac{1}{n}$ Here $a_n = \sin \frac{1}{n}$ mono. ↓ing & $\lim a_n = 0$

By L-test $\sum a_n$ is cgt.

$\sum |a_n| = \sum \sin \frac{1}{n}$ is div. $\because$ By comparison test, taking $b_n = 1/n$

$\lim \frac{a_n}{b_n} = \lim \frac{\sin 1/n}{1/n} = 1 \neq 0$ finite & $\sum b_n = \sum \frac{1}{n}$ is div.

Hence $\sum (-1)^n \sin \frac{1}{n}$ is cond. cgt.

Eg 2 For what values of p, $\sum 1 - \frac{1}{2^p} + \frac{1}{3^p} - \frac{1}{4^p} \dots$ is abs & cond. cgt?

$\sum a_n = \sum (-1)^n \frac{1}{n^p}$ $\therefore \sum |a_n| = \sum \frac{1}{n^p}$ is cgt for $p > 1$

& div for $p \leq 1$

Hence series $\sum a_n$ is abs. cgt for $p > 1$

& cond. cgt for $0 < p \leq 1$